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Defense-in-Depth Trust Layer for Multi-Agent LLMs

Built on AMD Instinct MI300X · Apache-2.0 + MIT · Solo dev · UNT, Argentina

Paper v2.0.1 · Zenodo DOI 10.5281/zenodo.20114594

5 slides · ~3 minute pitch · TechEx 2026

Two failure modes nobody catches together

Obvious attacks

Lobster Trap (Veea, MIT) 

- Prompt injection
- Credential exposure
- PII leakage
- Data exfiltration
- Role impersonation

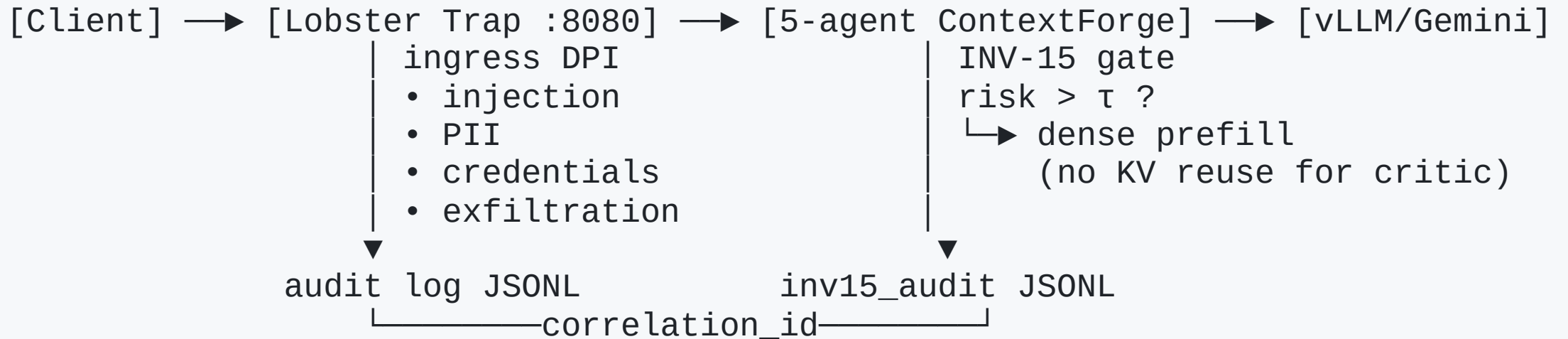
Silent drift

ContextForge INV-15 (Apache-2.0) 

- JCR drops **8-23pt** under KV reuse
- Judge agrees with cached output
- Compliance has no audit trail

Liang et al. 2026,
arXiv:2601.08343

Our answer: INV-15 + Lobster Trap composed



Layer 1 = perimeter (deterministic).

Layer 2 = behavioral (formal invariant).

Audit trails correlated by request_id.

Measured (real MI300X, not extrapolation)

3.55x

INT4 KV reduction
constant 4K → 262K context

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Lobster Trap adversarial
suite PASS on our policy

0 / 1,210

INV-15 violations on
exhaustive sweep

Δ 0.23

JCR drop *prevented*
(apohara_on 1.00 vs off 0.77)

Why it wins where others don't

	Apohara×LT	TrustLayer	Cursor	Lovable	GSD-2
Perimeter DPI (LT integration)	✓	✓	✗	✗	✗
Formal invariant for KV reuse	✓ NV-15	✗	✗	✗	✗
Paper with public DOI	✓ Zenodo	✗	✗	✗	✗
AUDIT.md honesty log	✓ public	✗	✗	✗	✗
Hardware					